Coordinate Geometry — Revision Summary

Lines, conics, and the algebra of geometric loci

1 Straight lines

1.1 Equations of a line

Form	Equation
General	$ax + by + c = 0$ (slope $-a/b$ when $b \neq 0$)
Slope-intercept	$y = mx + c$ ($m =$ slope, $c = y$-intercept)
Point-slope	$y - y_1 = m(x - x_1)$
Two-point	$\dfrac{y - y_1}{y_2 - y_1} = \dfrac{x - x_1}{x_2 - x_1}$
Intercept	$\dfrac{x}{a} + \dfrac{y}{b} = 1$ ($a, b = x$- and y-intercepts)
Normal	$x \cos\alpha + y \sin\alpha = p$ ($p =$ perpendicular distance from origin, $\alpha =$ angle with x-axis)
Parametric	$\dfrac{x - x_1}{\cos\theta} = \dfrac{y - y_1}{\sin\theta} = r$ ($r =$ signed distance from (x_1, y_1))

1.2 Distance, angle, and perpendicularity

- **Distance from** (x_0, y_0) **to** $ax + by + c = 0$**:** $\dfrac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- **Distance between parallel lines** $ax + by + c_1 = 0$, $ax + by + c_2 = 0$: $\dfrac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$.
- **Angle between lines** with slopes m_1, m_2: $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right|$.
- **Parallel:** $m_1 = m_2$. **Perpendicular:** $m_1 m_2 = -1$ (or one vertical, other horizontal).

1.3 Foot of perpendicular and image

For point $P = (x_1, y_1)$ and line $L : ax + by + c = 0$:

Foot of perpendicular. (x', y') given by

$$\frac{x' - x_1}{a} = \frac{y' - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}.$$

Image (reflection) of P in L. (x'', y'') given by

$$\frac{x'' - x_1}{a} = \frac{y'' - y_1}{b} = -\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}.$$

(Same formula as foot, with the factor 2 instead of 1 — image is twice as far as the foot from P.)

1.4 Family of lines and concurrency

Family through intersection of $L_1 = 0$ and $L_2 = 0$. Every line through the point of intersection has the form $L_1 + \lambda L_2 = 0$ for some $\lambda \in \mathbb{R}$ (plus $L_2 = 0$ itself).

Concurrency of three lines. $a_1 x + b_1 y + c_1 = 0$, etc., are concurrent iff

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

1.5 Position of points and angle bisectors

Same/opposite sides. Two points (x_1, y_1), (x_2, y_2) lie on the *same side* of $ax + by + c = 0$ iff $(ax_1 + by_1 + c)(ax_2 + by_2 + c) > 0$.

Angle bisectors. The two angle bisectors of $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$ are

$$\frac{a_1 x + b_1 y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2 x + b_2 y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

The $+$ sign gives the bisector of the angle containing the origin if $c_1 c_2 > 0$ (with the convention $c_1, c_2 > 0$ after sign-normalising).

2 Pair of straight lines

2.1 Homogeneous second-degree equation

Theorem. *$ax^2 + 2hxy + by^2 = 0$ represents a pair of straight lines through the origin iff $h^2 \geq ab$. The two lines are real and distinct if $h^2 > ab$, real and coincident if $h^2 = ab$, imaginary if $h^2 < ab$.*

Slopes. If the two lines are $y = m_1 x$ and $y = m_2 x$:

$$m_1 + m_2 = -\frac{2h}{b}, \qquad m_1 m_2 = \frac{a}{b}.$$

Angle between them.

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a+b|}.$$

Perpendicular pair. $a + b = 0$. **Coincident pair:** $h^2 = ab$.

2.2 General second-degree equation as pair of lines

Theorem. *$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines iff*

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \quad \textit{and} \quad h^2 \geq ab.$$

Point of intersection of the pair. Solve the system $\partial S/\partial x = 0$, $\partial S/\partial y = 0$, i.e., $ax + hy + g = 0$, $hx + by + f = 0$.

Joint equation trick. The pair of lines joining the origin to the intersection of $L : y = mx + c$ ($c \neq 0$) and conic $S(x, y) = 0$ is obtained by *homogenising* S with L: replace any constant by $\left(\frac{y - mx}{c}\right)^k$ to make every term degree 2.

3 Circles

3.1 Standard equations

Centre $(0, 0)$, radius r	$x^2 + y^2 = r^2$
Centre (h, k), radius r	$(x - h)^2 + (y - k)^2 = r^2$
General form	$x^2 + y^2 + 2gx + 2fy + c = 0$
centre	$(-g, -f)$
radius	$\sqrt{g^2 + f^2 - c}$ (real iff $g^2 + f^2 \geq c$)
Diameter form (endpoints A, B)	$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$
Parametric	$(h + r\cos\theta, k + r\sin\theta)$

3.2 The $T = S_1$ family of equations

For circle $S \equiv x^2 + y^2 + 2gx + 2fy + c$, define

$$S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c \quad \text{(evaluation at } (x_1, y_1)\text{)},$$

$$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c.$$

Object	Equation
Tangent at (x_1, y_1) on the circle	$T = 0$
Chord of contact from external (x_1, y_1)	$T = 0$
Polar line of (x_1, y_1)	$T = 0$
Equation of chord with midpoint (x_1, y_1)	$T = S_1$
Pair of tangents from external (x_1, y_1)	$SS_1 = T^2$

3.3 Tangent and normal

Tangent in slope form. $y = mx \pm r\sqrt{1 + m^2}$ (to circle $x^2 + y^2 = r^2$).

Length of tangent. From external point (x_1, y_1) to circle $S = 0$: $\sqrt{S_1}$.

Normal. Always passes through the centre. So the normal at (x_1, y_1) is the line through (x_1, y_1) and the centre.

3.4 Power, radical axis, family

Power of a point. For point P and circle $S = 0$, $\text{pow}(P) = S_1$ (i.e., evaluate S at P). P is inside iff power < 0, on iff $= 0$, outside iff > 0.

Radical axis of two circles $S = 0$, $S' = 0$. The locus of points with equal power w.r.t. both circles: $S - S' = 0$. (For circles in general form, this simplifies to a linear equation.) The radical axis is perpendicular to the line of centres.

Radical centre. The point of intersection of the radical axes of three circles (taken in pairs) is the radical centre, which has equal power with respect to all three circles.

Family of circles.

- Through intersection of $S = 0$ and $S' = 0$: $S + \lambda S' = 0$.
- Through intersection of $S = 0$ and line $L = 0$: $S + \lambda L = 0$.
- Concentric to $S = 0$: $x^2 + y^2 + 2gx + 2fy + \lambda = 0$.

3.5 Two circles — positions and common tangents

For circles with centres C_1, C_2 and radii r_1, r_2, let $d = |C_1C_2|$:

Configuration	Common tangents
$d > r_1 + r_2$ (circles disjoint, external)	4 (2 direct + 2 transverse)
$d = r_1 + r_2$ (externally tangent)	3
$\lvert r_1 - r_2\rvert < d < r_1 + r_2$ (intersecting)	2 (direct only)
$d = \lvert r_1 - r_2\rvert$ (internally tangent)	1
$d < \lvert r_1 - r_2\rvert$ (one inside other)	0

Director circle. Locus of points from which two perpendicular tangents can be drawn to $x^2 + y^2 = r^2$: $x^2 + y^2 = 2r^2$.

4 Parabola

Standard form $y^2 = 4ax$ (opens rightward, $a > 0$).

4.1 Key elements

Vertex	$(0, 0)$
Focus	$(a, 0)$
Directrix	$x = -a$
Axis	$y = 0$ (the x-axis)
Latus rectum (length)	$4a$ (chord through focus perpendicular to axis)
Endpoints of latus rectum	$(a, \pm 2a)$
Parametric form	$(at^2, 2at)$
Eccentricity	$e = 1$

Other standard parabolas.

- $y^2 = -4ax$: opens leftward, focus $(-a, 0)$.
- $x^2 = 4ay$: opens upward, focus $(0, a)$.
- $x^2 = -4ay$: opens downward, focus $(0, -a)$.

4.2 Tangent and normal at (x_1, y_1)

For $y^2 = 4ax$:

$$\text{Tangent: } yy_1 = 2a(x + x_1), \qquad \text{Normal: } y - y_1 = -\frac{y_1}{2a}(x - x_1).$$

Parametric tangent at $(at^2, 2at)$. $ty = x + at^2$, slope $1/t$.

Parametric normal at $(at^2, 2at)$. $y + tx = 2at + at^3$, slope $-t$.

Tangent in slope form. $y = mx + a/m$ (touches at $(a/m^2, 2a/m)$).

4.3 Conormal points and subtangent/subnormal

Conormal points. In general, 3 normals can be drawn from a given point (h, k) to the parabola $y^2 = 4ax$. The roots m_1, m_2, m_3 of the resulting cubic equation satisfy $m_1 + m_2 + m_3 = 0$.

- **Sum of y-coordinates** of the feet of the normals is zero: $y_1 + y_2 + y_3 = 0$.

- The locus of the **centroid** of the conormal triangle lies on the axis of the parabola (the x-axis).

Subtangent and subnormal. At any point (x_1, y_1) on the parabola: length of subtangent is $2x_1$ (bisected at the vertex), and length of subnormal is $2a$ (constant).

4.4 Focal-chord and reflection properties

Focal distance. From point (x_1, y_1) on the parabola to focus: $\boxed{x_1 + a}$ (= distance to the directrix). This is the defining property.

Focal chord through t_1, t_2. Endpoints $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ form a focal chord iff $t_1t_2 = -1$.

Reflection property. A ray parallel to the axis reflects off the parabola through the focus. Equivalently: the tangent at any point bisects the angle between the focal chord and the axis-parallel line through that point. (Underlies parabolic reflectors.)

Pedal property. The foot of the perpendicular drawn from the focus to any tangent lies on the tangent at the vertex.

Director circle (Orthoptic locus). The directrix $x = -a$ is the locus of intersection of perpendicular tangents.

$T = S_1$ analogues for parabola. For $S \equiv y^2 - 4ax$:

- Tangent at (x_1, y_1): $T \equiv yy_1 - 2a(x + x_1) = 0$.
- Chord with midpoint (x_1, y_1): $T = S_1$.
- Pair of tangents from external (x_1, y_1): $SS_1 = T^2$.

5 Ellipse

Standard form $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, with $a > b > 0$ (major axis along x-axis).

5.1 Key elements

Centre	$(0, 0)$
Vertices (major)	$(\pm a, 0)$. Co-vertices: $(0, \pm b)$
Foci	$(\pm ae, 0)$ where $e = \sqrt{1 - b^2/a^2}$
Directrices	$x = \pm a/e$
Eccentricity	$0 < e < 1$
Latus rectum (length)	$2b^2/a$
Parametric form	$(a\cos\theta, b\sin\theta)$, $\theta \in [0, 2\pi)$
Auxiliary circle	$x^2 + y^2 = a^2$
Director circle	$x^2 + y^2 = a^2 + b^2$

5.2 Defining properties (focal-distance)

Sum-of-focal-distances. For any point P on the ellipse with foci F_1, F_2:

$$|PF_1| + |PF_2| = 2a.$$

This is the geometric definition of an ellipse.

Focal distance from (x_1, y_1). To focus $(ae, 0)$: $a - ex_1$. To focus $(-ae, 0)$: $a + ex_1$. Sum: $2a$ (consistent with the defining property).

Focal-directrix property. Distance to focus / distance to corresponding directrix = e for every point on the curve.

5.3 Tangent and normal at (x_1, y_1)

$$\text{Tangent: } \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1, \qquad \text{Normal: } \frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2.$$

Parametric tangent at $(a\cos\theta, b\sin\theta)$. $\dfrac{x\cos\theta}{a} + \dfrac{y\sin\theta}{b} = 1$.

Tangent in slope form. $y = mx \pm \sqrt{a^2m^2 + b^2}$.

Reflection property. A ray from one focus reflects off the ellipse to the other focus. ("Whispering gallery.")

5.4 Conjugate diameters and advanced properties

Conjugate diameters. Two diameters $y = m_1x$ and $y = m_2x$ of the ellipse are conjugate iff $m_1m_2 = -b^2/a^2$. Each bisects all chords parallel to the other.

Property at endpoints of conjugate diameters. If P and Q are endpoints of conjugate semi-diameters with eccentric angles α and $\alpha + \pi/2$, then:

- $|CP|^2 + |CQ|^2 = a^2 + b^2$ (constant).
- Area of parallelogram with conjugate diameters as sides $= 4ab$.

Conormal points. In general, 4 normals can be drawn from a point to an ellipse. The sum of the eccentric angles of the four conormal points is an *odd* multiple of π, i.e., $\alpha + \beta + \gamma + \delta \equiv \pi \pmod{2\pi}$.

Concyclic points. If a circle intersects the ellipse in four distinct points, the sum of their eccentric angles is an *even* multiple of π, i.e., $\alpha + \beta + \gamma + \delta \equiv 0 \pmod{2\pi}$.

Auxiliary-circle correspondence. A point $P(a\cos\theta, b\sin\theta)$ on the ellipse corresponds directly to the point $Q(a\cos\theta, a\sin\theta)$ on the auxiliary circle, sharing the same eccentric angle θ.

Reciprocal of square radii. For any two mutually perpendicular chords CP and CQ drawn through the centre C:

$$\frac{1}{|CP|^2} + \frac{1}{|CQ|^2} = \frac{1}{a^2} + \frac{1}{b^2}.$$

Locus of mid-point of a focal chord. The mid-points of focal chords drawn through $(ae, 0)$ lie on the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{ex}{a}$.

$T = S_1$ analogues for ellipse. With $S \equiv \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - 1$, $T \equiv \dfrac{xx_1}{a^2} + \dfrac{yy_1}{b^2} - 1$:

- Chord with midpoint (x_1, y_1): $T = S_1$.
- Chord of contact from external (x_1, y_1): $T = 0$.
- Pair of tangents from external (x_1, y_1): $SS_1 = T^2$.

6 Hyperbola

Standard form $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$.

6.1 Key elements

Centre	$(0, 0)$
Vertices	$(\pm a, 0)$
Foci	$(\pm ae, 0)$ where $e = \sqrt{1 + b^2/a^2}$
Directrices	$x = \pm a/e$
Eccentricity	$e > 1$
Latus rectum (length)	$2b^2/a$
Asymptotes	$y = \pm \frac{b}{a}x$ (combined: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$)
Parametric form	$(a\sec\theta, b\tan\theta)$
Conjugate hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ (foci on y-axis)

6.2 Defining properties

Difference-of-focal-distances. For any point P on the hyperbola: $\big||PF_1| - |PF_2|\big| = 2a$. Geometric definition.

Focal distance. To $(ae, 0)$: $|ex_1 - a|$. To $(-ae, 0)$: $|ex_1 + a|$. Difference: $2a$.

Reflection property. A ray directed toward one focus reflects off the hyperbola in the direction of the other focus (i.e., as if it had come from the other focus). Used in Cassegrain telescopes.

6.3 Tangent and normal at (x_1, y_1)

$$\text{Tangent: } \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1. \qquad \text{Normal: } \frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2.$$

Tangent in slope form. $y = mx \pm \sqrt{a^2m^2 - b^2}$ (real iff $a^2m^2 \geq b^2$).

Director circle. $x^2 + y^2 = a^2 - b^2$. Real only when $a > b$ (otherwise no point has perpendicular tangents).

$T = S_1$ analogues. With $S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$, $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$: same equations as ellipse case ($T = 0$ for chord of contact, $T = S_1$ for chord with given midpoint, $SS_1 = T^2$ for pair of tangents).

6.4 Asymptotes and conjugate diameters

Asymptote-tangent midpoint property. The point of contact of any tangent is the midpoint of the tangent segment intercepted between the two asymptotes.

Constant area property. The area of the triangle formed by any tangent and the two asymptotes is constant, equal to ab.

Conjugate diameters. Two diameters $y = m_1 x$ and $y = m_2 x$ are conjugate if $m_1 m_2 = b^2/a^2$.

Reflection in asymptotes. A hyperbola is its own reflection when mirrored across its transverse or conjugate axes, and bounded geometrically by the asymptote cone.

6.5 Rectangular hyperbola $xy = c^2$

The rectangular hyperbola has $a = b$, hence $e = \sqrt{2}$ and asymptotes perpendicular. Rotating by 45° gives the standard form $xy = c^2$.

Asymptotes	$x = 0$ and $y = 0$ (the coordinate axes)
Parametric form	$(ct, c/t)$, $t \neq 0$
Tangent at $(ct, c/t)$	$x + t^2 y = 2ct$, equivalently $\frac{x}{ct} + \frac{ty}{c} = 2$
Normal at $(ct, c/t)$	$t^3 x - ty = c(t^4 - 1)$
Foci	$(c\sqrt{2}, c\sqrt{2})$ and $(-c\sqrt{2}, -c\sqrt{2})$
Conormal points	For four conormal points, $t_1 t_2 t_3 t_4 = -1$

7 Loci and standard locus problems

The general principle: write coordinates of the moving point in terms of a parameter, eliminate the parameter, and simplify.

Geometric condition	Locus
Equidistant from two points A, B	Perpendicular bisector of AB
Distance from a fixed point is constant	Circle
Sum of distances from two points = const	Ellipse (those points as foci)
|Difference of distances| = const	Hyperbola (those points as foci)
Distance from point = distance from line	Parabola
Ratio dist-to-point / dist-to-line = e	Conic with eccentricity e
$\lvert PA\rvert/\lvert PB\rvert = k$ ($k \neq 1$, A, B fixed)	Apollonius circle
$\angle APB$ = const, A, B fixed	Arc of a circle through A, B
$\angle APB = 90°$, A, B fixed	Circle with AB as diameter
Foot of perp from focus on tangent (parabola)	Tangent at vertex
Foot of perp from focus on tangent (ellipse, hyperbola)	Auxiliary circle
Mid-points of parallel chords	A diameter of the conic
Perpendicular tangents (parabola)	Directrix
Perpendicular tangents (ellipse)	$x^2 + y^2 = a^2 + b^2$
Perpendicular tangents (hyperbola)	$x^2 + y^2 = a^2 - b^2$

Pole and polar. For a conic $S = 0$ and point $P = (x_1, y_1)$:

- The polar of P is the line $T = 0$.
- If P is outside, the polar is the chord of contact of tangents from P.
- If P is on the conic, the polar is the tangent at P.
- Pole-polar is a duality: if Q lies on the polar of P, then P lies on the polar of Q.

8 Quick-reference cheat sheet

Pattern in the problem	Tool / trick to try first
Lines	
Distance from point to line	$\lvert ax_0 + by_0 + c\rvert/\sqrt{a^2+b^2}$
Angle between two lines	$\tan\theta = \lvert(m_1 - m_2)/(1 + m_1 m_2)\rvert$
Image of a point in a line	"2× foot of perpendicular" formula
Concurrency of three lines	Determinant $= 0$
Family through two-line intersection	$L_1 + \lambda L_2 = 0$
Pair of lines	
$ax^2 + 2hxy + by^2 = 0$: when a pair?	$h^2 \geq ab$
Angle between pair	$\tan\theta = 2\sqrt{h^2 - ab}/\lvert a + b\rvert$
Pair perpendicular	$a + b = 0$
General second-degree pair condition	$\Delta = 0$
Pair of lines from origin to conic-line intersection	Homogenise
Circles	
Length of tangent from external point	$\sqrt{S_1}$
Power of a point	S_1
Pair of tangents from external point	$SS_1 = T^2$
Radical axis of two circles	$S - S' = 0$
Family through intersection of S, S'	$S + \lambda S' = 0$
Locus of perpendicular tangents (circle)	Director circle $x^2 + y^2 = 2r^2$
Parabola	
Focal distance from point on $y^2 = 4ax$	$x_1 + a$
Focal chord parameter relation	$t_1 t_2 = -1$
Tangent at $(at^2, 2at)$	$ty = x + at^2$, slope $1/t$
Tangent in slope form to $y^2 = 4ax$	$y = mx + a/m$
Locus of perpendicular tangents (parabola)	The directrix
Reflection (axis-parallel ray)	Reflects through focus
Ellipse and hyperbola	
Sum/difference of focal distances	$2a$
Focal distance, ellipse	$a \mp ex_1$
Focal distance, hyperbola	$\lvert ex_1 \mp a\rvert$
Tangent in slope form (ellipse)	$y = mx \pm \sqrt{a^2m^2 + b^2}$
Tangent in slope form (hyperbola)	$y = mx \pm \sqrt{a^2m^2 - b^2}$
Conjugate diameters (ellipse)	$m_1 m_2 = -b^2/a^2$
Director circle (ellipse)	$x^2 + y^2 = a^2 + b^2$
Auxiliary circle	$x^2 + y^2 = a^2$
Reflection (ellipse)	Focus to focus
Reflection (hyperbola)	Toward one focus, away from the other
Asymptotes of $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$	$y = \pm(b/a)x$
Rectangular hyperbola	$xy = c^2$, parametric $(ct, c/t)$
Universal $T = S_1$ family (every conic)	
Tangent at (x_1, y_1) on the curve	$T = 0$
Chord of contact from external (x_1, y_1)	$T = 0$
Polar of (x_1, y_1)	$T = 0$
Chord with midpoint (x_1, y_1)	$T = S_1$
Pair of tangents from external (x_1, y_1)	$SS_1 = T^2$

Three universal moves: (1) *Build T and S_1 first* — the entries $T = 0$, $T = S_1$, $SS_1 = T^2$ cover almost every chord/tangent/pole problem on any conic. (2) *Parametrise to attack loci* — write coordinates as a function of one parameter, eliminate it. (3) *Use the focal/directrix definition* — many problems collapse to one-line statements about focal distances.